Supporting Information:

Exploring Ligand-to-Metal Charge-transfer States in the Photo-Ferrioxalate System using Excited-State Specific Optimization

Lan Nguyen Tran*,†,‡ and Eric Neuscamman*,¶,§

†*Department of Physics, International University, Ho Chi Minh City, Vietnam*
‡*Vietnam National University, Ho Chi Minh City, Vietnam*
¶*Department of Chemistry, University of California, Berkeley, California, 94720, USA*
§*Chemical Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA, 94720, USA*
E-mail: tnlan@hcmiu.edu.vn; eneuscamman@berkeley.edu

1. Various experimental evidence on the prompt mechanism of the ferrioxalate photolysis

In 2008, Pozdnyakov and coworkers [1] used optical spectroscopy and nanosecond flash photolysis (Nd:YAG laser, 355 nm, pulse duration 5 ns, mean energy 5 mJ/pulse) to study the photolysis of ferrioxalate. They carried out the experiment with a wide range of parameters. They found that the primary photochemical process is the intramolecular electron transfer from the ligand to Fe(III) ion with the formation of a primary radical complex $[(C_2O_4)_2Fe^{II}(C_2O_4^{\bullet})]^{3-}$. They proposed a kinetic scheme for FeIII(C2O4)33- photolysis to support their conclusion. This study was the first to show evidence of the prompt mechanism of ferrioxalate photolysis. However, the photochemical intermediates were not revealed.

In 2015, Ogi and coworkers [2] performed time-resolved X-ray absorption spectroscopy using an X-ray free electron laser and a synchronized ultraviolet laser. They observed a red shift of the Fe K-edge and the spectral feature of the product, indicating that Fe(III) is, upon photo-excitation, immediately reduced to Fe(II) due to the ligand-to-metal charge transfer (LMCT). Based on geometry optimization for the LMCT state using time-dependent density functional theory (TD-DFT), the authors claimed that the primary process is the dissociation of the C—C bond, followed by the elongation of the Fe—O bond. The donor oxalate ligand then dissociates from Fe(II) to form CO_2^- and CO_2 spices.

In 2017, Mangiante and coworkers [3] employed pump/probe mid-infrared transient absorption spectroscopy to study ferrioxalate photolysis. They found that the intramolecular charge transfer from oxalate to iron occurs on a sub-picosecond time scale. Based on the appearance of a new vibrational absorption band and theoretical simulation, they stated that within 40 ps after the charge transfer, the oxidized oxalate dissociates to form free CO_2 and a species $CO_2^{\bullet-}$. They proposed that the LMCT generates an unstable oxalate anion that dissociates rapidly, likely on the timescale of a single C—C vibration. But, which bond, Fe—O or C—C, dissociates first was unclear.

In the same year, O'Neil and coworkers [4] employed a tabletop apparatus based on a laser plasma X-ray source and an array of cryogenic microcalorimeter X-ray detectors to measure a transient X-ray absorption spectrum during the photolysis of ferrioxalate. They observed the Fe K edge move to lower energies and the amplitude of the extended X-ray absorption fine structure reduce, supporting the intramolecular charge transfer from oxalate to iron center. They showed evidence of Fe—O bond elongation and provided quantitative limits on the Fe—O bond length change. However, these limits are small (0.14 ± 0.06 Å).

In 2018, Straub and coworkers [5] employed femtosecond mid-infrared spectroscopy combined with ultrafast laser photolysis to provide the missing information in the photochemical process of ferrioxalate. The authors suggested that by 1 ns, one oxalate ligand has entirely fragmented into a singly negatively charged carbon dioxide radical and a neutral CO2 molecule. In 2021, Longetti and coworkers [6] investigated the photoinduced dynamics of aqueous ferrioxalate using ultrafast liquid phase photoelectron spectroscopy. Upon the LMCT excitation, the excited molecule dissociates in <140 fs into the pair of CO_2 and $[(C_2O_4)_2Fe^{II}(C_2O_4^{\bullet})]^{3-}$ fragment.

2. XYZ coordinates of optimized geometries:

a. Ground-state equilibrium:

Fe	1.307060105001	2.485789995435	-0.587735556152
O	0.283436224784	4.132722915431	-1.230745064622
O	2.709862031871	3.968548726043	-0.504091306845
C	0.888864691312	5.238081777853	-1.294126375541
C	2.379118795613	5.137591706506	-0.845239339708
O	3.087745765159	6.112317430714	-0.847489884920
O	0.425098020476	6.290982443785	-1.652694769033
O	0.693922094534	2.668476522438	1.353167571591
O	2.464331868669	1.116210417367	0.393216132109
O	0.969764050290	1.939824170528	3.404578402088
O	2.926677109864	0.215027586337	2.340474426221
C	1.234899193813	1.939088564834	2.228964780820
C	2.319545759182	0.985133298998	1.639940218563
O	1.864455434904	1.854929961983	-2.449666575929
O	-0.174126084072	1.171929745545	-1.092495714652
C	1.135546458261	1.016398073925	-3.052855549254
C	-0.118899043774	0.593025454557	-2.215186732351
O	1.323530957426	0.549713315597	-4.149528339300
O	-0.906941465014	-0.206441220132	-2.657495318017

b. Bound LMCT equilibrium:

- *Optimized using SS-CASSCF(7e,6o)*

Fe	1.936424	1.189184	-0.182057
O	0.764414	2.442592	-1.510515
O	3.156222	2.846189	-0.610914
C	1.209186	3.592409	-1.760677
C	2.682486	3.821229	-1.253843
O	3.245473	4.867133	-1.495581
O	0.631190	4.493086	-2.330600
O	0.908109	0.969119	1.634459
O	3.243338	0.001808	1.053898
O	0.980419	0.142434	3.680015
O	3.478173	-0.881150	3.066848
C	1.447998	0.328694	2.577028
C	2.869484	-0.254097	2.222284
O	2.106680	0.388117	-4.294214
O	1.429141	-0.448133	-1.573838
C	1.041038	0.203827	-3.810020

C	0.706467	-0.638228	-2.545475
O	-0.047380	0.620480	-4.446223
O	-0.197719	-1.428649	-2.733803

- *Optimized using SS-CASSCF(7e,12o)*

Fe	1.921358	1.192667	-0.200900
O	0.823925	2.426449	-1.600840
O	3.125427	2.879931	-0.518720
C	1.264057	3.584589	-1.822691
C	2.686549	3.844227	-1.200886
O	3.246696	4.900032	-1.402181
O	0.715417	4.469335	-2.443249
O	0.914083	0.953843	1.630189
O	3.250407	0.013367	1.005454
O	1.025041	0.109033	3.666648
O	3.521011	-0.887273	3.005473
C	1.473510	0.310154	2.557656
C	2.894465	-0.256650	2.176664
O	2.149019	0.302391	-4.183143
O	1.279436	-0.458760	-1.503582
C	1.047018	0.183216	-3.764761
C	0.590480	-0.609463	-2.506613
O	0.023721	0.633209	-4.481659
O	-0.360478	-1.334257	-2.726088

- *Optimized using TD-DFT/B3LYP:*

Fe	1.365109	2.607533	-0.434375
O	0.270011	4.187192	-1.247682
O	2.768024	4.084449	-0.431078
C	0.904030	5.289703	-1.398055
C	2.415314	5.230669	-0.905851
O	3.137864	6.225525	-0.995990
O	0.457499	6.343032	-1.866474
O	0.719386	2.723404	1.494038
O	2.485057	1.081168	0.443835
O	0.924127	1.792577	3.526664
O	2.824505	0.033743	2.407019
C	1.213470	1.876001	2.331262
C	2.282008	0.884603	1.692817
O	1.896533	1.877585	-2.529457
O	-0.222712	1.100531	-1.068460
C	1.139730	1.128557	-3.164442
C	-0.131810	0.472750	-2.133962
O	1.094823	0.757583	-4.331773
O	-0.729077	-0.487256	-2.607045

c. Stretched C—C (2.2 Å) geometry:

Fe	1.879944	2.136776	-0.112964
O	1.035186	3.963533	-0.863193
O	3.493641	3.458781	-0.248863
C	1.787010	4.964756	-0.950898
C	3.291176	4.654177	-0.600852
O	4.122906	5.530437	-0.680549
O	1.472834	6.093698	-1.264475
O	0.656195	1.616152	1.501366
O	2.768249	0.360111	0.704351
O	0.384810	0.257764	3.220818
O	2.647665	-1.094482	2.365473
C	0.968427	0.632311	2.228370
C	2.256481	-0.131686	1.739182
O	2.107414	0.258819	-4.705506
O	1.269720	1.105992	-1.958465
C	1.063553	-0.209443	-4.414284
C	0.326145	0.829558	-2.616235
O	0.238610	-1.006001	-4.713742
O	-0.834354	0.832852	-2.753889

d. Transition state:

Optimized using SA(3)-CASSCF(11e,8o)

Fe	0.000000000000	0.000000000000	0.000000000000
O	0.000000000000	0.000000000000	2.324739555700
C	0.000000000000	0.448439009700	3.444727028000
O	0.644699943100	1.226784613600	4.088234205300
C	-1.145237135900	-0.282683101100	4.434625478900
O	-0.865208117800	-1.135352307200	5.211616491500
O	-2.241812916700	0.298413286500	4.187392191400
C	-2.369546680200	0.940647430400	-1.443276028300
O	-3.319824328000	1.045171509200	-2.185680019100
O	-2.454712441800	3.228759728100	-0.760952390800
O	1.960929840600	0.243807423700	-0.696815818200
O	0.757840054500	-1.990783664500	-0.154000185800
O	3.788349239900	-0.788046316200	-1.364274460100
O	2.496621829700	-3.185885914000	-0.793034741200
C	2.659631480300	-0.772371179000	-0.928667170300
C	1.917009195500	-2.142728597300	-0.599248758700
O	-0.927400391800	1.917586168000	0.148608088500
O	-1.679998564400	-0.087337903700	-1.244005454100
C	-1.890840554700	2.172079373500	-0.615219670700

3. Active spaces for SS-CASSCF(7e,6o):

Active orbitals of SS-CASSCF(7e,6o) calculations for the ground state: one ligand orbital and five Fe 3d orbitals. These orbitals are also the initial guess for SS-CASSCF(7e,6o) for the LMCT state at the ground-state geometry. The ligand orbital was selected based on the work of Ogi et al. [2]. Upon optimization, the (7e,6o) active orbitals look very similar to the lone pair and Fe 3d orbitals shown in Figure 3 of the main text.

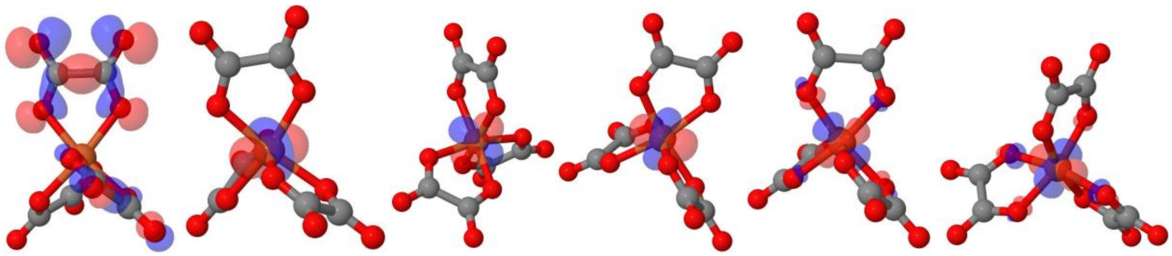

Figure S1

4. Active space for SA-CASSCF(11e,8o):

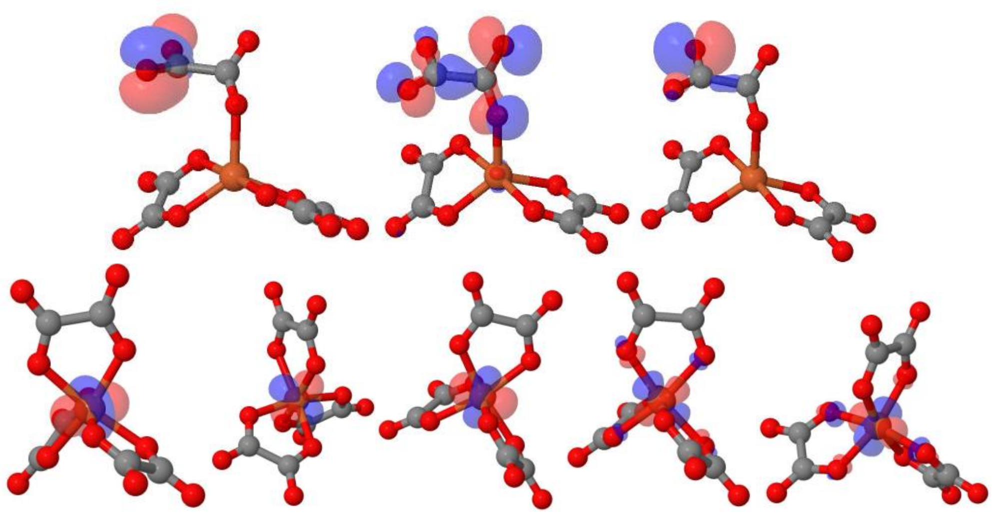

Figure S2

References:

[1] Pozdnyakov, I. P.; Kel, O. V.; Plyusnin, V. F.; Grivin, V. P.; Bazhin, N. M. New insight into photochemistry of ferrioxalate. *J. Phys. Chem. A* **2008**, *112*, 8316—8322.

[2] Ogi, Y.; Obara, Y.; Katayama, T.; Suzuki, Y.-I.; Liu, S.; Bartlett, N.-M.; Kurahashi, N.; Karashima, S.; Togashi, T.; Inubushi, Y. et al. Ultraviolet photochemical reaction of [Fe (III)(C2O4) 3] 3- in aqueous solutions studied by femtosecond time-resolved X-ray absorp- tion spectroscopy using an X-ray free electron laser. *Struct. Dyn.* **2015**, *2*, 034901.

[3] Mangiante, D. M.; Schaller, R. D.; Zarzycki, P.; Banfield, J. F.; Gilbert, B. Mechanism of ferric oxalate photolysis. *ACS Earth Space Chem.* **2017**, *1*, 270—276.

[4] O'Neil, G. C.; Miaja-Avila, L.; Joe, Y. I.; Alpert, B. K.; Balasubramanian, M.; Sagar, D.; Doriese, W.; Fowler, J. W.; Fullagar, W. K.; Chen, N. et al. Ultrafast time-resolved X-ray absorption spectroscopy of ferrioxalate photolysis with a laser plasma X-ray source and microcalorimeter array. *J. Phys. Chem. Lett.* **2017**, *8*, 1099—1104.

[5] Straub, S.; Bru¨nker, P.; Lindner, J.; Vo¨hringer, P. An Iron Complex with a Bent, O-Coordinated CO2 Ligand Discovered by Femtosecond Mid-Infrared Spectroscopy. *Angew. Chem.* **2018**, *130*, 5094–5099.

[6] Longetti, L.; Barillot, T.; Puppin, M.; Ojeda, J.; Poletto, L.; van Mourik, F.; Arrell, C.; Chergui, M. Ultrafast photoelectron spectroscopy of photoexcited aqueous ferrioxalate. *Phys. Chem. Chem. Phys.* **2021**, *23*, 25308—25316.